

Azure Devops

- **Introduction**

- Why Devops?
- Implementing Devops
- Azure DevOps tools

- **Azure Devops services – Azure Boards**

- Azure DevOps processes
- Permissions in Azure DevOps
- Using charts in Azure DevOps

- **Source Code Versioning**

- What is GIT?
- Hosting your code centrally
- GitHub main branch naming change
- Azure Repos
- GitFlow
- Azure Repos - Forking a repository

- **Continuous Integration**

- What is Continuous Integration
- The build process
- Understanding Azure Pipelines
- Starting with Azure Pipelines
- Azure Pipelines - Editing our YAML configuration
- Azure Pipelines and GitHub
- Azure Pipelines and GitHub Authentication

- Jenkins - Azure Repos
- Jenkins - Integrating with GitHub
- Different testing processes
- Making code changes after using an Azure Pipeline
- Azure Pipelines - Unit Testing

● Building Your Infrastructure

- Azure Automation
- PowerShell Desired State Configuration – Extension
- Azure Resource Manager templates
- What is Terraform

● Continuous Delivery

- Continuous Delivery
- Understanding Azure Release Pipelines
- Deploying docker
- Deploying to Kubernetes
- Azure Build Pipelines - Azure Kubernetes
- Azure Release Pipelines – Kubernetes
- What is Helm
- Working with Helm charts
- Azure Pipelines - System defined variable
- Azure Traffic Manage
- Blue Green Deployments

- **Develop a site Reliability Engineering and Instrumentation Strategy**

- Understanding Site Reliability
- Azure Virtual Machine Scale Sets
- Review on monitoring and logging
- Diagnostics for your resources
- Alerts in Azure Monitor
- Dynamic Alerts
- Monitoring your virtual machines
- What is Application Insights
- What is Prometheus

- **Test plans, Package Management and Security**

- Azure Artifacts
- Azure Security Overview
- Multi-Factor Authentication